

FAST – National University of Computer and Emerging Sciences, Karachi.
FAST School of Computing
Assignment # 1, Fall 2026
CS3001 – Computer Networks

Note: The assignment consists of TWO sections, and both sections are compulsory. Late submissions will not be accepted.

Submission Deadline: 15 Sep, 2026

Section (A) DNS QUERY RESOLUTION USING SOCKET PROGRAMMING

1. Objective

[100 Marks]

You will implement a DNS query program to understand DNS as an application-layer protocol and how a client communicates with DNS servers to resolve a hostname into an IP address. This is primarily a programming exercise applying the DNS concepts covered in class: the distributed hierarchical database (root, TLD, and authoritative servers), local DNS servers, DNS caching/TTL, and basic resource records.

2. Programming Task

Write a program that:

- Accepts a domain name from the user (e.g., www.example.com).
- Sends a DNS query to a specified DNS server using a UDP socket.
- Receives and parses the DNS response.
- Displays, at minimum: Transaction ID, query name, query type, response status/relevant flags, the answer record(s), the resolved IPv4 address for an A record (if available), and the TTL (if available).
- The DNS server IP address should be provided at runtime. The program must use the standard DNS port 53.
- Handles invalid domain names and unsuccessful responses (e.g., NXDOMAIN, timeout) gracefully, without crashing.
- Allows the user to look up multiple domain names in one run, without restarting the program.

3. Important Requirements

- Use socket programming with UDP; queries must go to the standard DNS port 53.
- Do NOT use a high-level resolver call (e.g., `gethostbyname()` or equivalent) as the core implementation you must construct and send the DNS query and parse the raw response yourself.
- Correctly build/parse the basic DNS message structure (see Section 4).
- Handle request timeouts and malformed/invalid responses.
- The program must work with any domain name supplied at runtime. Do not hard-code a single domain.
- Code should be modular, readable, and properly commented.

4. Minimum DNS Message Handling

Understand and handle the basic DNS message layout: Header → Question → Answer. When constructing the query, set an appropriate Transaction ID, flags (e.g., recursion desired), a question count of 1, and encode the domain name and query type (A) correctly. From the response, extract the header fields, the question section, and the answer record(s) needed to report the resolved address and TTL. You are not required to implement a full production-grade or recursive DNS resolver parsing the Header, Question, and Answer sections for an A-record lookup is sufficient.

5. Testing / Demonstration

Demonstrate your program with at least THREE different domain names: one common .com domain, one .edu or .org domain, and one domain of your choice. Provide screenshots showing the program execution, domain name entered, DNS server used, parsed DNS response, resolved IP address, and TTL.

6. Supporting Links

- RFC 1035: Domain Names: Implementation, Specification and message format reference: <https://www.rfc-editor.org/rfc/rfc1035>
- Python socket programming: official documentation: <https://docs.python.org/3/library/socket.html>
- Beej's Guide to Network Programming Using Internet Sockets: <https://beej.us/guide/bgnet/>

Section (B) Numerical / Problem-Solving Questions

Answer all FIVE questions. Show all calculations; final answers without working will receive partial credit only. It must be handwritten. **[20 Marks]**

Q1. DNS Resolution Time [2 + 2 = 4 Marks]

A client's local DNS server has no cached information and must resolve `www.netlearn.edu` using iterative queries. The round-trip query + response times are: Root DNS server = 60 ms, TLD DNS server = 50 ms, Authoritative DNS server = 40 ms.

- Calculate the total DNS resolution time.
- If the responses of both the Root and TLD servers are already cached locally (so the local DNS server queries only the authoritative server), calculate the time saved compared to part (a).

Q2. DNS Caching and TTL [4 Marks]

A DNS record for `cdn.streamhub.com` is first resolved and cached at 10:00:00 AM with a TTL of 300 seconds. A full (uncached) resolution takes 150 ms, while a cache lookup takes 2 ms.

- Is the cached entry still valid for a query made at 10:03:00 AM? If valid, how much time is saved by using the cache instead of a full resolution?
- Is the cached entry still valid for a query made at 10:07:00 AM? Justify your answer.

Q3. Transmission Delay [4 Marks]

A file of size 12 MB is sent from a host across two links in sequence: Link 1 operates at 20 Mbps and Link 2 operates at 100 Mbps. (1 MB = 8 Megabits.) Ignore propagation, queuing, and processing delay. Calculate the transmission delay on Link 1, on Link 2, and the total transmission delay for the file to cross both links.

Q4. End-to-End Delay [4 Marks]

A packet of size 8000 bits travels from a server to a client across three links: Link 1 (40 Mbps, propagation delay 20 ms), Link 2 (80 Mbps, propagation delay 30 ms), and Link 3 (500 Mbps, propagation delay 5 ms). Assume no queuing delay and no processing delay at any node. Calculate the total end-to-end delay experienced by the packet from server to client.

Q5. Web Cache / Hit Ratio [4 Marks]

A web server receives 5000 requests per hour, with each requested object averaging 400 Kb in size. A caching proxy is introduced, achieving a cache hit ratio of 75%.

- Compute the total traffic per hour without caching.
- Compute the traffic per hour that actually reaches the origin server after caching is introduced.
- Compute the amount of data traffic saved per hour due to caching.

Submission Guidelines

- Individual assignment.
- Any language allowed, Python recommended.
- Submit your source code and screenshots of a working execution. Make a consolidated PDF of both tasks [Programming_Screenshots {Section A} and solved numericals {Section B} with all the necessary steps].
- A short demonstration/viva will be conducted.
- Include your student name, student ID, and section in the submission.
- Plagiarism in any form will result in a straight "F".